Francesco Marrocco

M.S. ECE with 2+ years of professional experience in hardware/chip design and characterization
Worcester Polytechnic Institute
Boston, MA | fmmarrocco@wpi.edu | 201-509-0310
LinkedIn: https://www.linkedin.com/in/fmmarrocco/ GitHub: https://github.com/Vision314

EDUCATION

Worcester Polytechnic Institute — Worcester, MA
M.S. Electrical & Computer Engineering — GPA 3.9 May 2026
Thesis: *Design and Characterization of a 4MS/s 8-bit Flash ADC in 180nm CMOS for Dual-Channel Image Sensor Readout*

B.S. Electrical & Computer Engineering and Minor in Data Science — GPA 3.8 May 2025
High Distinction & Dean's List 2023–2025

TECHNICAL SKILLS

Circuit Rework
Hands-on experience with tricky PCB alterations that require patience & skill

Hardware Testing
Strong understanding of testing equipment to properly characterize a DUT

Documentation & Writing
LaTeX in Emacs; technical writing for academic papers and documentation

High-Level Programming
Python – pandas, pyserial, pyvisa, numpy, matplotlib, scipy

Hardware Description Languages
Verilog/SV – timing considerations, FSMs, serial comms (SPI, I2C, UART)

Embedded Programming
C/C++ with microcontrollers like MSP430 and Arduino

Altium
Schematic & layout experience with DFM, DFT, and avoiding R-EMI issues

Simulation
LTSpice – analog simulation
Xilinx Vivado – testbenches for FPGA's

Other Tools
Cadence Virtuoso – simulation & IC layout; **Fusion360** – 3D printing

EXPERIENCE

Medical Power Supply Design Engineer
GlobTek Inc. — *Northvale, NJ* — June 2025 – Present
Skills Used: Altium, Python, EMI debugging, LTSpice, LaTeX, PCB rework and hardware testing

- Design and test power supplies for medical-grade applications, with careful consideration of IC and topology selection, thermal management, EMI compliance, size constraints, transformer optimization, and manufacturing capabilities.
- Developed Python automation tools that modernize legacy test equipment, replacing many hours of tedious lab work with immediate results.

Independent Researcher - Microelectronic Characterization
WPI CHIPS Design Studio — *Worcester, MA* — August 2024 – Present
https://wpi-chips-design-studio.github.io/wpi-chips-lab/
Skills Used: Altium, Python, Cadence Virtuoso, LaTeX, chip testing, C/C++, Verilog/SystemVerilog, Xilinx Vivado

- Led by Dr. Suat Ay, the CHIPS Studio focuses on solving problems with custom integrated circuits.
- Completed my master's thesis with funding provided by the CHIPS Studio and CEMiD (Center for Education of Microchip Designers @ UCLA) where I designed, taped-out, and characterized a Flash ADC on 180nm CMOS.
- Currently a volunteer researcher leading validation and testing for upcoming research.

INVOLVEMENT

First Place Robert H. Grant Invention Award, WPI
Skills Used: C/C++, Fusion360, Altium — March 2024 – May 2024

- Built a food spoilage detection device ("Freshness Finder") by measuring the methane emitted from rotting food; delivered a fully functioning prototype in three weeks: Perfboard assembly, Arduino programming, 3D-printed enclosure

Microelectronics Tutor & Electronics Design Mentor, WPI — August 2024 – October 2024
Worcester, MA

- Lab instructor and tutor for topics surrounding fundamental semiconductor physics, including in-depth analysis of diode behavior, MOSFET operation, and op-amp characteristics.

DIII Men's Varsity Rowing, WPI — August 2021 – May 2025

- Strong work ethic demonstrated by 4 years of 25+ hours per week of training along with rigorous academics